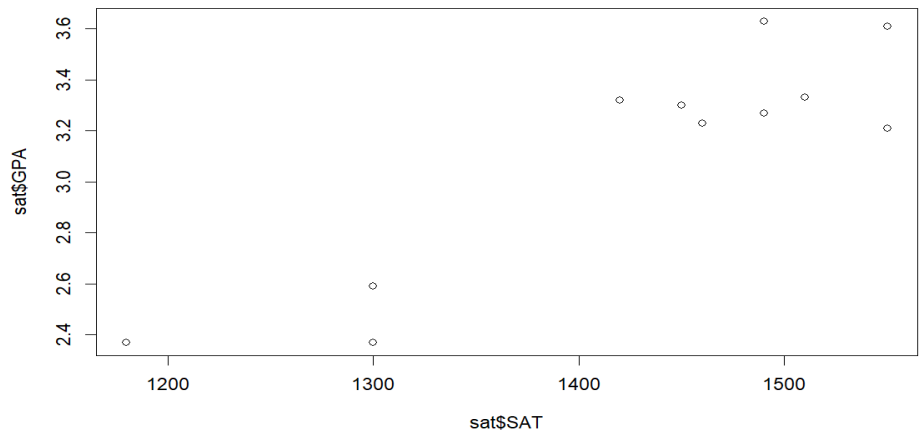
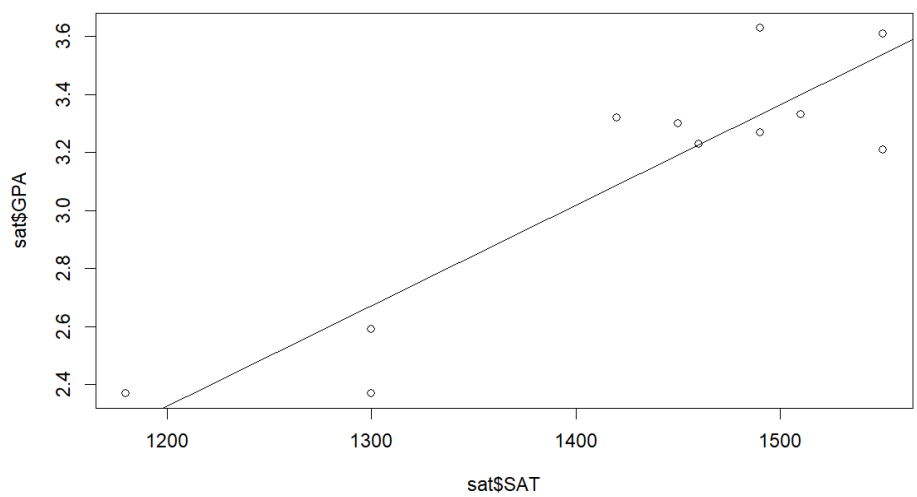
A. Is SAT score a good predictor of GPA?

Yes, SAT score is a good predictor of GPA. The scatter plot shows a positive relationship, and the p-value (0.000152) is very small, indicating statistical significance. Additionally, the R^2 value (0.8122) shows that SAT explains a large portion of the variation in GPA. $t \approx 5.7$, p-value = 0.000152 < 0.05.





B. Interpretation of slope

The slope indicates that for each 1-point increase in SAT score, GPA increases by approximately 0.00346 points on average.

C. Hypothesis test (β = 0)

H_0: $\beta = 0$
H_a: $\beta \neq 0$

$\alpha = 0.05$

p-value = 0.000152

Since p-value < 0.05, we reject the null hypothesis.

Conclusion: SAT score is significantly related to GPA.

D. Why is the intercept negative?

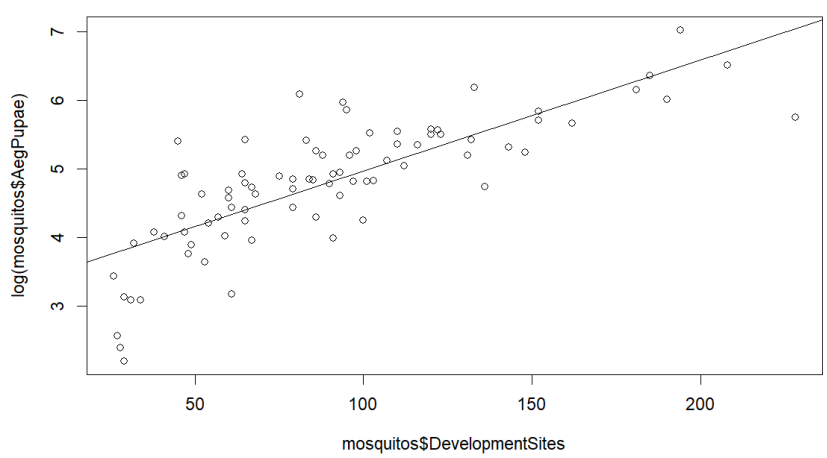
The intercept is negative because it represents GPA when SAT = 0, which is not realistic. Therefore, it has no meaningful real-world interpretation.

E. Percentage explained

Approximately **81.22%** of the variation in GPA is explained by SAT scores.

F. Interpretation of slope (linear mosquito model)

For each additional development site, the number of mosquito pupae increases by approximately **2.80 pupae** on average.



G. Percent change in β

= [(2.7959 − 2.3536) / 2.7959] × 100
≈ 15.82%

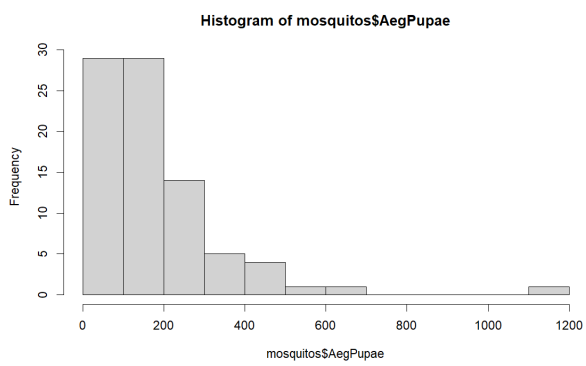
Removing the outlier reduces the estimated effect of development sites on pupae by about 15.82%.

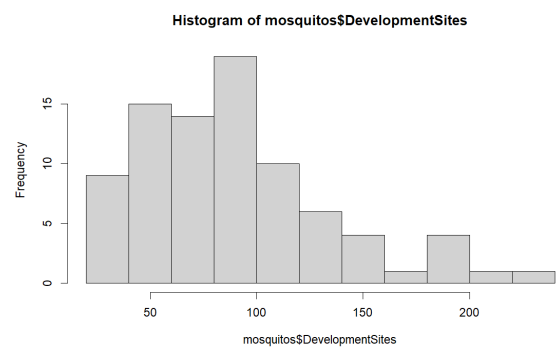
H. Why does t-value increase?

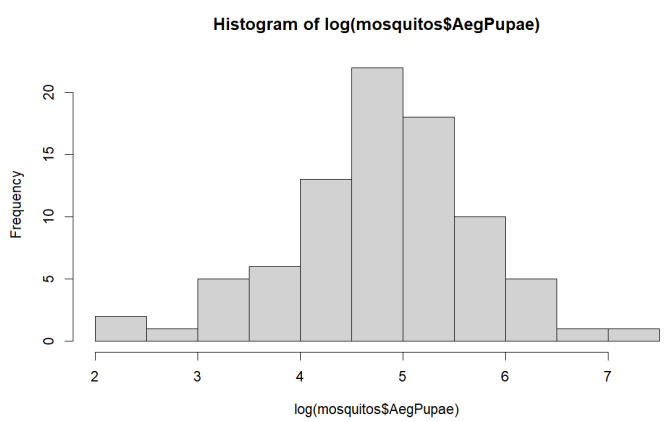
The t-value increases because removing the outlier reduces variability and improves model fit, making the estimate more precise.

I. Histogram comparison

The raw pupae data is highly right-skewed. After log transformation, the distribution becomes more symmetric and closer to normal.

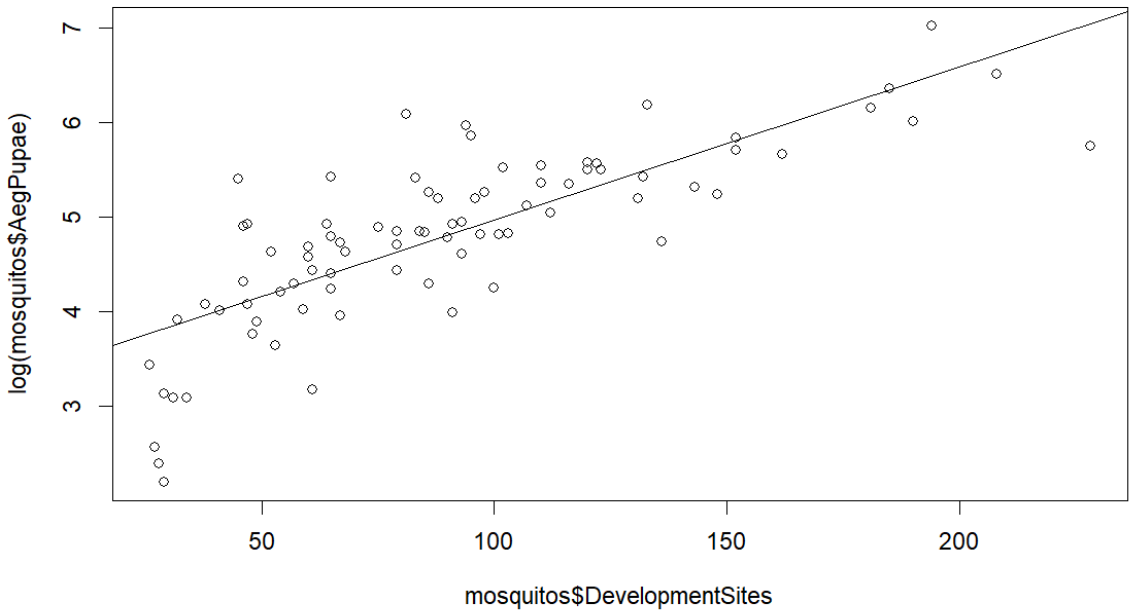






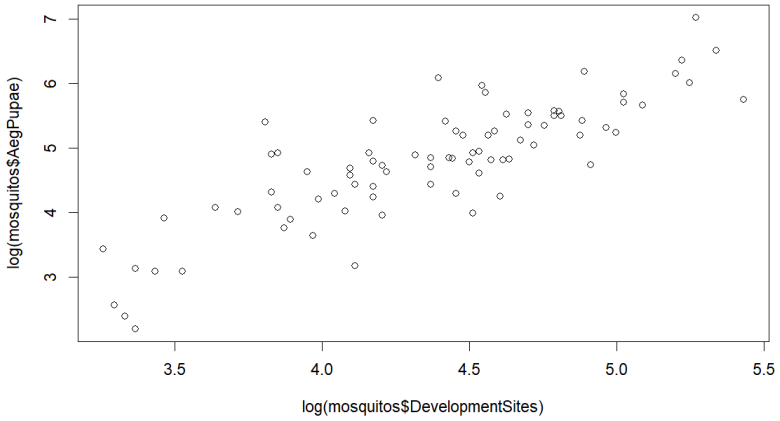
J. Interpretation of slope (log model)

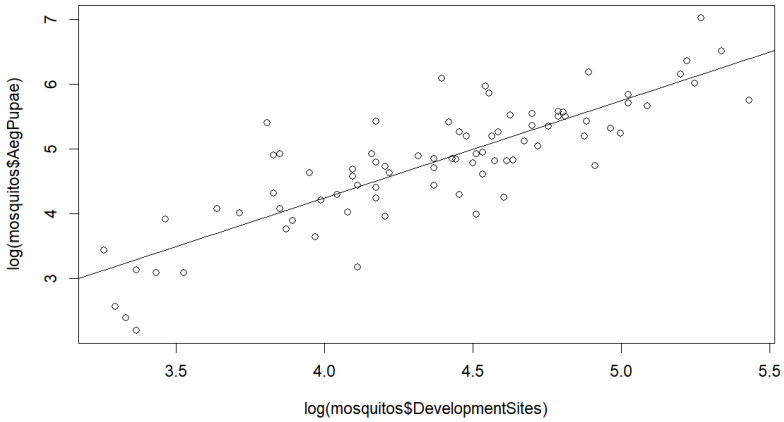
For each additional development site, the number of pupae increases by approximately **1.62%** on average.



K. Interpretation of slope (log-log model)

The slope represents elasticity. A 1% increase in development sites leads to approximately a **1.50% increase** in pupae.





L. Hypothesis test ($\rho^2 = 0$)

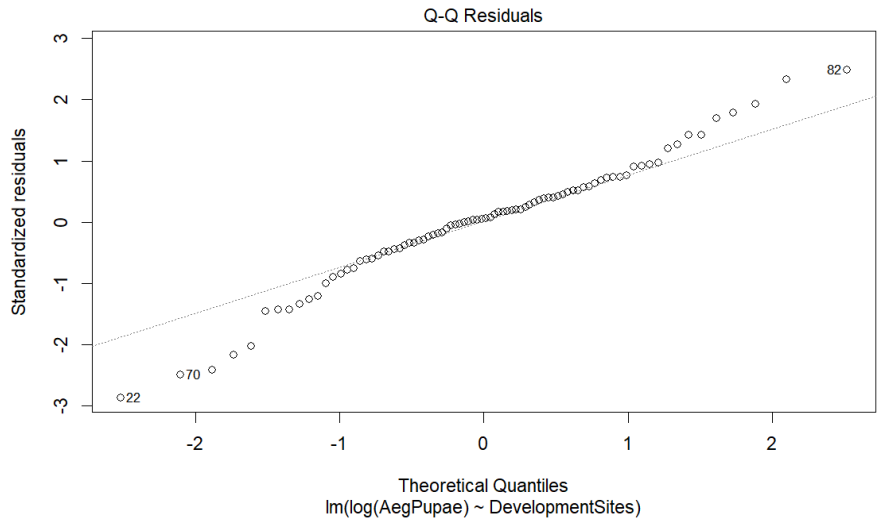
H_0: $R^2 = 0$
H_a: $R^2 > 0$

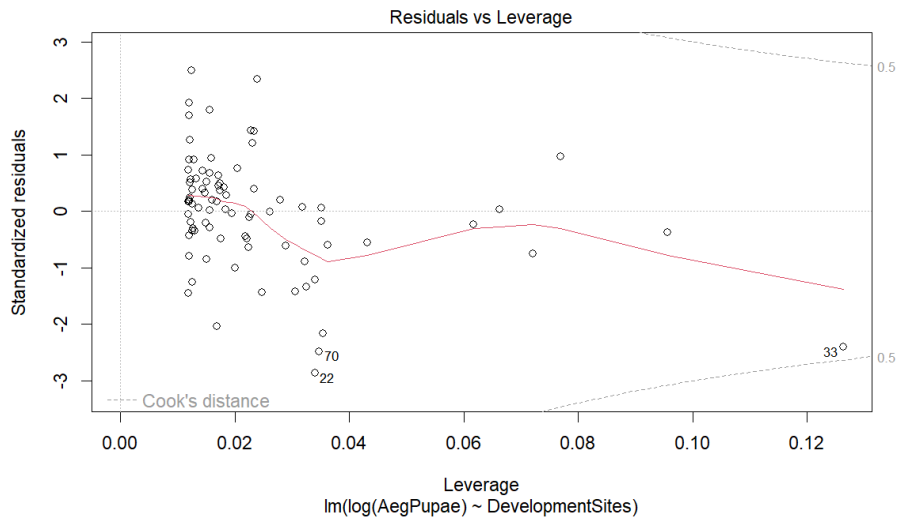
$\alpha = 0.05$

p-value < 2.2e-16

Since p-value < 0.05, we reject the null hypothesis.

Conclusion: The model explains a significant portion of variation in mosquito pupae.


Q-Q Residuals
Standardized residuals
Theoretical Quantiles
lm(log(AegPupae) ~ DevelopmentSites)
82
70
22


Residuals vs Leverage
Standardized residuals
Leverage
lm(log(AegPupae) ~ DevelopmentSites)
Cook's distance
0.5
0.5
70
22
33